Project Information

Project Name:	Math Careers and Salaries
Your Name:	Kristine Steele
Date:	1/20/2026

Data Details

Data Domain:	Careers, Education, Labor Market
Data Source:	Custom dataset created for academic analysis
Number of Columns:	6
Number of Records:	25

Attribute Summary Table

Column Name	Description
JobTitle	Name of the math-related career
Industry	Sector employing the role
MedianSalary	Typical annual salary
EducationLevel	Minimum degree required
GrowthRate	Projected job growth
Region	Geographic area of employment

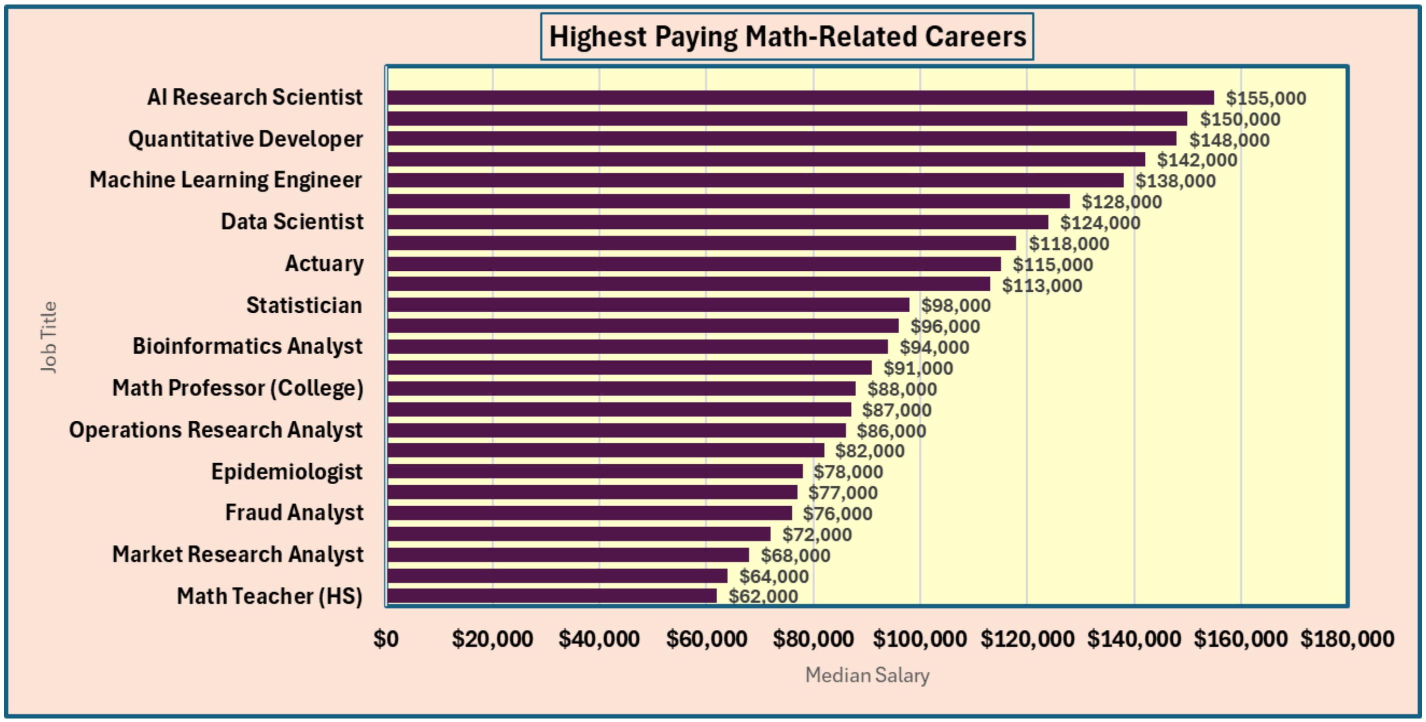
Assumptions

1. This data set reflects realistic incomes and growth projections for math-related careers across major U.S. regions.
2. Job titles and industries are indicitive of common career directions for people with strong mathematical

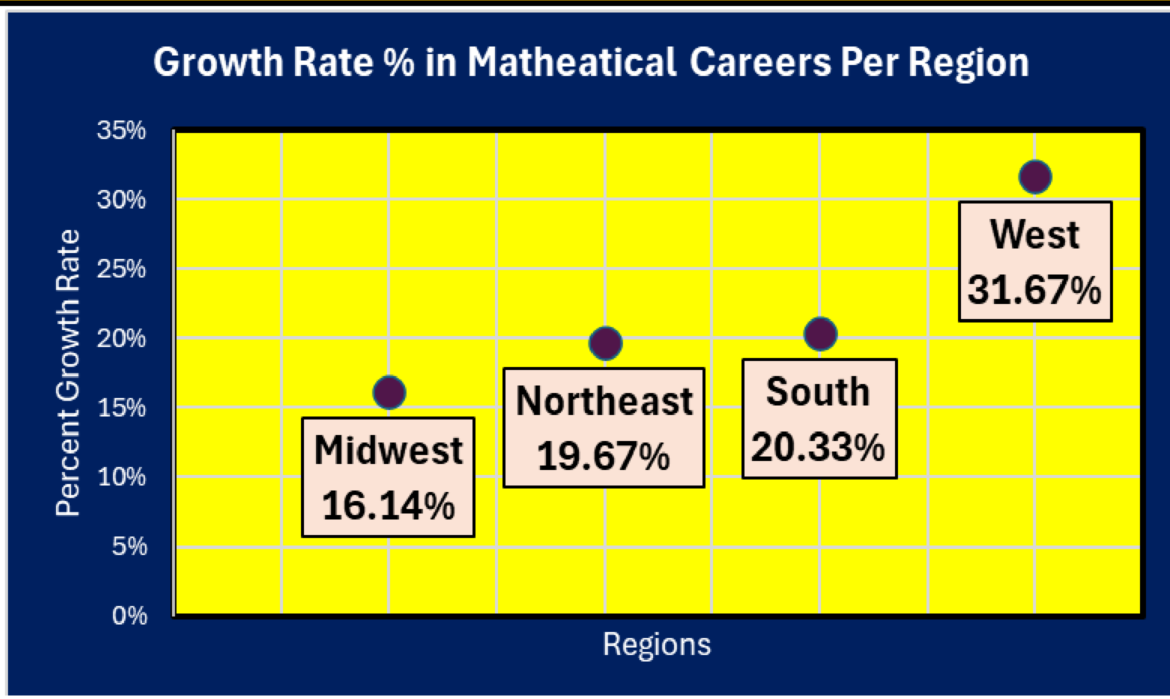
Goal 1:	Highest Paying Careers in Math Related Fields
Goal:	"Which math-related careers have the highest median salaries?"
Attributes (Multiple):	Job Title, Median Salary

Attribute Summary Table:	
Column Name	Description
Job Title	Name of the math-related career
Median Salary	Typical annual salary for the role

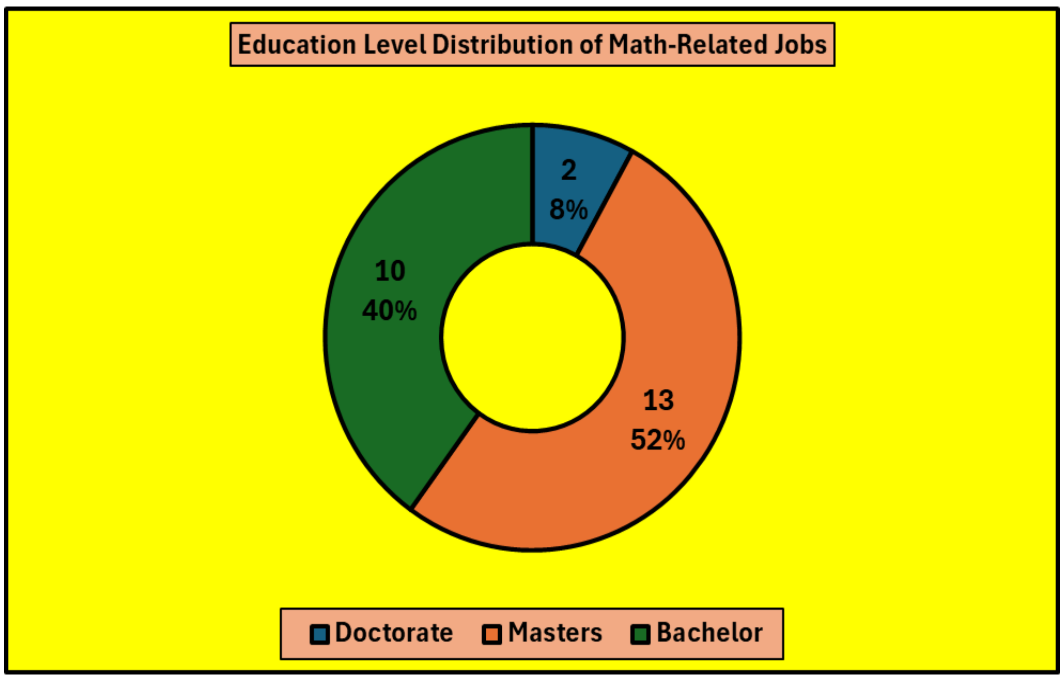
Selected Chart Type:	Bar Chart
Why this Chart Works:	This horizontal bar chart clearly compares salary values across all math-related careers, making it easy to identify the highest-paying roles.
Data Story:	"The bar chart shows clear differences in median salaries across math-related careers. AI Research Scientist is highest at $155,000, while Math Teacher (HS) is lowest at $62,000. High-paying roles like Financial Mathematician ($150,000) and Quantitative Developer ($148,000) sit near the top, reflecting advanced technical demands."



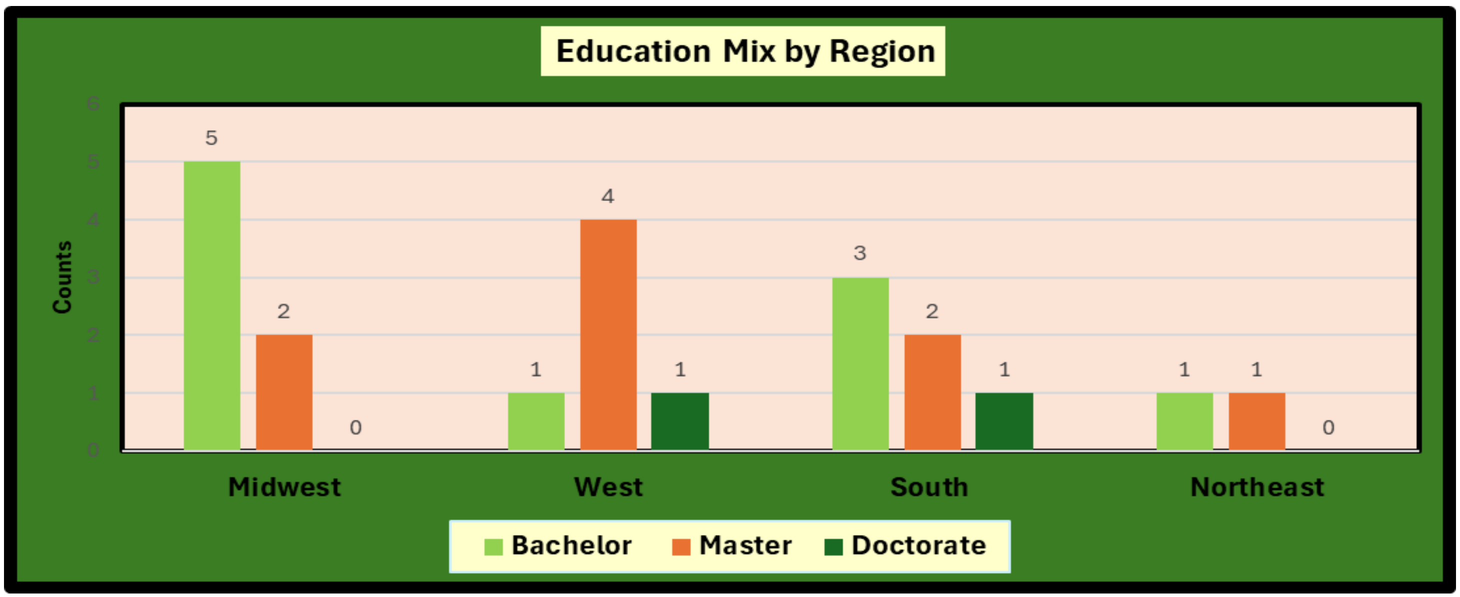
Goal 2:	Regional Growth Rate Comparison
Goal:	"Which region has the highest average growth rate for mathematic based jobs?"
Attributes (Multiple)	Region, GrowthRate
Attribute Summary Table:	
Column Name	Description
Region	Geographic region of the job
Growth Rate	Projected growth rate for mathematical jobs (percent)
Selected Chart Type:	Dot plot
Why this Chart Works:	A dot plot is ideal here because it compares a small number of regional averages clearly on a single numeric scale without visual clutter.
Data Story:	Average projected job growth varies by region. The West has the highest average growth at 31.67%, indicating stronger expected demand for these math-related roles. The South (20.33%) and Northeast (19.67%) are mid-range, while the Midwest is lowest at 16.14%.



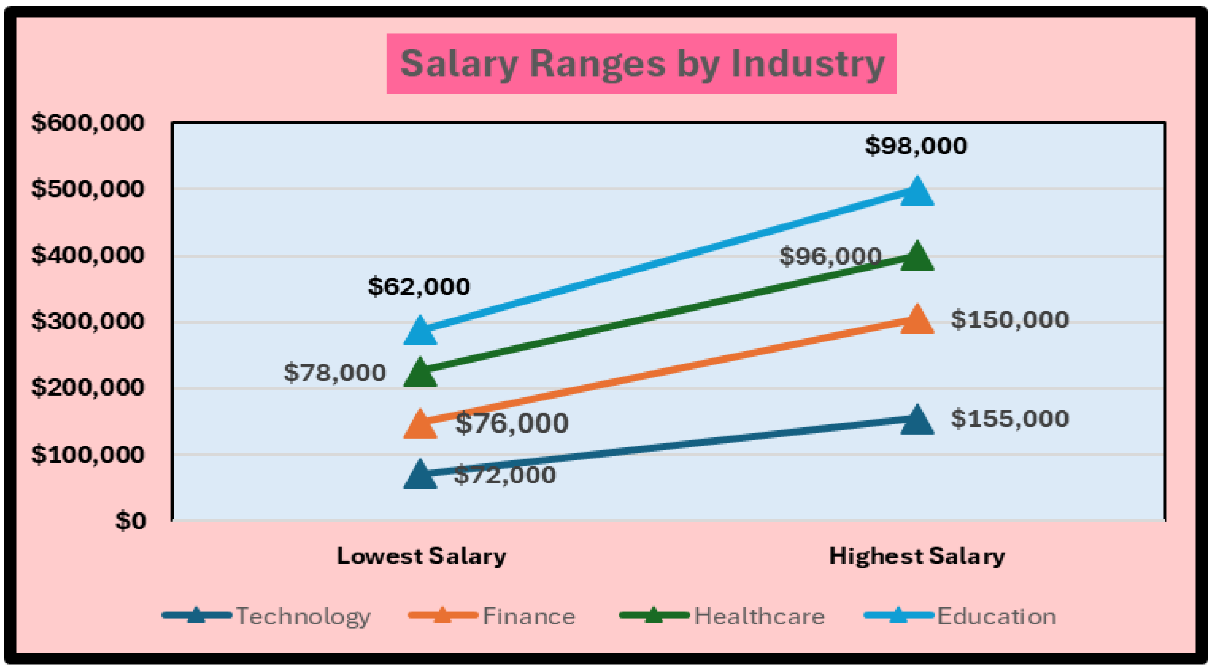
Goal 3:	Education Level Distribution
Goal:	How are math-related jobs distributed by education level?
Attributes (Multiple):	Education Level, Count
Attribute Summary Table:	
Donut Portions	Description
Education Level	Highest degree required for the role
Count	Number of jobs in each edcuation level category
Selected Chart Type:	Donut Chart
Why this Chart Works:	A donut chart works well because it shows how each education level contributes to the overall distribution of jobs at a glance.
Data Story:	The donut chart shows the distribution of math-related jobs by education level. Master's degrees account for 13 of 25 roles (52%), Bachelor's for 10 of 25 (40%), and Doctorates for 2 of 25 (8%). This indicates most positions require graduate-level preparation, while doctoral training is comparatively rare.



Goal 4:	Education Mix by Region
Goal:	How does the education mix (Bachelor, Master, Doctorate) differ by region for math-related jobs?
Attributes (Multiple):	Region, Education Level, Count
Attribute Summary Table:	
Columns	Description
Region	Geographic Region of Job
Education Level	Highest Degree Required
Count	Number of jobs in each education level category
Selected Chart Type:	Side-by-side bar (clustered column)
Data Story:	**The clustered bar chart shows distinct education patterns by region. The West and Northeast are master's-heavy (West 4, Northeast 5), while the Midwest has the strongest bachelor's presence (5). Doctorate roles are rare overall and appear only in the West and South (1 each), suggesting advanced research roles are concentrated in fewer regions.**



Goal 5:	Salary Ranges by Industry (Slope Chart)
Goal:	How do salary ranges (lowest vs highest roles) differ across industries?
Attributes (Multiple)	Industry, Median Salary
Attribute Summary Table:	
Column	**Description**
Industry	**Industry category used for comparison**
Median Salary	**Annual median salary for roles in each industry**
Selected Chart Type:	Slope Chart
Why this Chart Works:	A slope chart clearly shows the size of change between two endpoints (lowest to highest) for each industry.
Data Story:	Technology has the widest salary range ($72k–$155k), with Finance close behind ($76k–$150k), indicating larger pay dispersion in those industries. Healthcare ($78k–$96k) and Education ($62k–$88k) have tighter ranges, suggesting more consistent pay levels. Overall, higher-tech industries show greater upside.



JobTitle	Industry	MedianSal	EducationLev	GrowthI	Region
Math Teacher (HS)	Education	62000	Bachelor	5%	Midwest
Survey Researcher	Government	64000	Master	4%	Midwest
Market Research Analyst	Marketing	68000	Bachelor	19%	South
Data Analyst	Technology	72000	Bachelor	23%	Midwest
Fraud Analyst	Finance	76000	Bachelor	14%	Midwest
Logistician	Manufacturing	77000	Bachelor	18%	Midwest
Epidemiologist	Healthcare	78000	Master	26%	South
Financial Analyst	Finance	82000	Bachelor	13%	Northeast
Operations Research Analyst	Consulting	86000	Master	28%	Midwest
Business Intelligence Analyst	Technology	87000	Bachelor	21%	Midwest
Math Professor (College)	Education	88000	Doctorate	9%	South
Risk Analyst	Insurance	91000	Bachelor	18%	South
Bioinformatics Analyst	Healthcare	94000	Master	27%	West
Clinical Data Manager	Healthcare	96000	Bachelor	17%	South
Statistician	Government	98000	Master	33%	South
Economist	Government	113000	Master	6%	Northeast
Actuary	Insurance	115000	Master	24%	Northeast
Data Engineer	Technology	118000	Bachelor	31%	West
Data Scientist	Technology	124000	Master	35%	West
Cryptographer	Security	128000	Master	30%	Northeast
Machine Learning Engineer	Technology	138000	Master	37%	West
Quantitative Analyst	Finance	142000	Master	22%	West
Quantitative Developer	Finance	148000	Master	25%	Northeast
Financial Mathematician	Finance	150000	Master	20%	Northeast
AI Research Scientist	Technology	155000	Doctorate	38%	West

Region	Growth Rate
Midwest	0.1614
Northeast	0.1967
South	0.2033
West	0.3167

EducationLevel	Count
Doctorate	2
Masters	13
Bachelor	10

Region	Bachelor	Master	Doctorate
Midwest	5	2	0
West	1	4	1
South	3	2	1
Northeast	1	1	0

Industry	Lowest Salary	Highest Salary
Technology	72000	2E+05
Finance	76000	2E+05
Healthcare	78000	96000
Education	62000	98000